



**SWAMINARAYAN
UNIVERSITY**
THE RESERVOIR OF ALL KNOWLEDGE

Faculty of Engineering (DEGREE)

Semester –7TH BE

Course Title – Introduction to Machine Learning

Course Code – BE104705

Types of Course -Program Core

Teaching and Examination Scheme –

Hours per Week			Course Credits	Examination Marks				150
L	T	P		Theory Marks		Tutorial /Practical		
				E	M	I	V	
3	0	2	4	60	40	20	30	

Detailed Syllabus –

Unit No.	Content	No. of Hours	% of Weightage
1.	Introduction to Machine Learning: Overview of Human Learning and Machine Learning, Types of Machine Learning, Applications of Machine Learning, Tools and Technology for Machine Learning .	2	4
2.	Preparing to Model: Machine Learning activities, Types of data in Machine Learning, Structures of data, Data quality and remediation, Data Pre-Processing: Dimensionality reduction, Feature subset selection.	4	8
3.	Modeling and Evaluation: Selecting a Model: Predictive/Descriptive, Training a Model for supervised learning, model representation and interpretability, Evaluating performance of a model, Improving performance of a model.	5	11
4.	Basics of Feature Engineering: Feature and Feature Engineering, Feature transformation: Construction and extraction, Feature subset selection : Issues in high-dimensional data,	3	6

	key drivers, measure and overall process		
5.	Overview of Probability: Statistical tools in Machine Learning, Concepts of probability, Random variables, Discrete distributions, Continuous distributions, Multiple random variables, Central limit theorem, Sampling distributions, Hypothesis testing, Monte Carlo Approximation	4	8
6.	Bayesian Concept Learning: Impotence of Bayesian methods, Bayesian theorem, Bayes' theorem and concept learning, Bayesian Belief Network	5	11
7	Supervised Learning: Classification and Regression: Supervised Learning, Classification Model, Learning steps, Classification algorithms, Regression, Regression algorithms	10	22
8	Unsupervised Learning: Supervised vs. Unsupervised Learning, Applications, Clustering, Association rules	6	13
9	Neural Network: Introduction to neural network, Biological and Artificial Neurons, Types of Activation functions, Implementation of ANN, Architecture, Learning process, Back propagation, Deep Learning	6	13
	Total	45	100

Course Outcome:

After Completion of the Course, Student will able to:

No	Course Outcomes	RBT level
CO1	Explore the fundamental issues and challenges in Machine Learning	Understand
CO2	including data and model selection and complexity	Apply
CO3	Appreciate the underlying mathematical relationships within and across	Apply
CO4	Machine Learning algorithms	Apply
CO5	Evaluate the various Supervised Learning algorithms using appropriate	Understand

References/Suggested Learning Resources:

Books:

- (a) Machine Learning, Saikat Dull, S. Chjandramouli, Das, Pearson
- (b) Machine Learning with Python for Everyone, Mark Fenner, Pearson
- (c) Machine Learning, Anuradha Srinivasaraghavan, Vincy Joseph, Wiley
- (d) Machine Learning with Python, U Dinesh Kumar Manaranjan Pradhan, Wiley
- (e) Python Machine Learning, Sebastian Raschka, Vahid Mirjalili, Packet Publishing

Open source software and website:

Library	Language	Description
TensorFlow	Python, C++	Developed by Google, widely used for both research and production. Great for deep learning.
PyTorch	Python	Developed by Facebook. Dynamic computation graph, intuitive, dominant in academia and research.
scikit-learn	Python	Built on NumPy/SciPy. Best for traditional ML algorithms (SVM, Random Forest, etc.).
JAX	Python	From Google. High-performance ML with composable function transformations and GPU/TPU support.
Keras	Python	High-level API for building and training deep learning models, now tightly integrated with TensorFlow.
ML.NET	C#	Microsoft's ML framework for .NET developers.
Shogun	C++ (with interfaces for Python, R, Java)	Focuses on kernel methods and SVMs.

Suggested Course Practical List:

1. To study and implement basic Python programming constructs used in machine learning applications.
2. To perform data acquisition, visualization, and exploratory data analysis using appropriate tools.
3. To implement data preprocessing techniques including handling missing values, normalization, and feature scaling.
4. To implement simple linear regression for prediction of continuous values.
5. To implement multiple linear regression and evaluate model performance.
6. To implement logistic regression for binary classification problems.
7. To implement k-Nearest Neighbors (k-NN) algorithm for classification tasks.
8. To implement decision tree algorithm for classification and prediction.
9. To implement Naïve Bayes classifier for probabilistic learning.

10. To implement Support Vector Machine (SVM) for classification tasks.
11. To perform clustering using k-Means algorithm for unsupervised learning.
12. To evaluate machine learning models using performance metrics such as accuracy, precision, recall, and F1-score.

List of Laboratory / Learning Resources Required

1. **Python (Anaconda / Jupyter Notebook)** – For implementation of machine learning algorithms.
2. **NumPy** – For numerical computations and matrix operations.
3. **Pandas** – For data manipulation and analysis.
4. **Matplotlib / Seaborn** – For data visualization and graphical representation.
5. **scikit-learn** – For implementing machine learning algorithms such as regression, classification, and clustering.
6. **TensorFlow / PyTorch** – For basic deep learning model implementation.
7. **Google Colab / Jupyter Notebook** – For cloud-based and interactive coding environment.
8. **Open Datasets (UCI / Kaggle)** – For training and testing machine learning models.

Suggested Project List

1. Study and implementation of basic Python libraries (NumPy, Pandas) for data handling.
2. Implementation of data preprocessing techniques including cleaning, normalization, and feature selection.
3. Development of a Linear Regression model for prediction of real-world data.
4. Implementation of Logistic Regression for classification problems.
5. Development of a k-Nearest Neighbors (k-NN) based classification system.
6. Implementation of Decision Tree model for classification and prediction tasks.
7. Development of a Naïve Bayes classifier for probabilistic learning.
8. Implementation of k-Means clustering for grouping unlabeled data.
9. Development of a simple machine learning application such as spam email detection or house price prediction.
10. Performance evaluation of machine learning models using accuracy, precision, recall, and F1-score.